DISTINCT POINTS IN THE SAME ORBIT UNDER A FINITE GROUPOID ARE NOT SPECIALIZATIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-groupoids.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Editorial note. Manual completeness read applied to the full 14-line proof: the specialization lifting, the no-specializations-in-finite-fibres step and the finiteness contradiction are all written out; nothing omitted or left to the reader.

Allowed standard prerequisites (editorial).

- Lifting specializations along quasi-compact closed arrows: specialization can be found after a base, giving a nontrivial specialization upstairs with prescribed image (Schemes chapter, Lemma schemes-lemma-quasi-compact-closed, canonical tag 01K9). cited at proof lines 2972-2975. Produces, from the assumed specialization u to u', a nontrivial specialization r to r' in R with s(r') = u', starting the infinite descent.

Source reference. Original lemma 01K9: sources/stacks-chow/tags.txt, source lines 2042–2042. Cited at proof lines 2972-2975: from the specialization u to u' it produces the nontrivial specialization r to r' with s(r') = u'.

Lemma 0.2. *Let (U, R, s, t, c) be a groupoid scheme. If s, t are finite, and $u, u' \in R$ are distinct points in the same orbit, then u' is not a specialization of u.*

Proof. Let $r \in R$ with $s(r) = u$ and $t(r) = u'$. If $u \rightsquigarrow u'$ then we can find a nontrivial specialization $r \rightsquigarrow r'$ with $s(r') = u'$, see Schemes, Lemma 01K9. Set $u'' = t(r')$. Note that $u'' \neq u'$ as there are no specializations in the fibres of a finite morphism. Hence we can continue and find a nontrivial specialization $r' \rightsquigarrow r''$ with $s(r'') = u''$, etc. This shows that the orbit of u contains an infinite sequence $u \rightsquigarrow u' \rightsquigarrow u'' \rightsquigarrow \ldots$ of specializations which is nonsense as the orbit $t(s^{-1}(\{u\}))$ is finite. □

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